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Design, Implementation and Verification of a Simulink-Based Digital Hearing Aid: A Signals and Systems Perspective

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A Report on

"Design, Implementation and Verification of a Simulink-Based Digital Hearing Aid: A Signals and Systems Perspective"

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CERTIFICATE

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This is to certify that the following students of B.Tech Semester – V, Division **EXCS A**, from Electronics and Computer Science have successfully completed the project titled **"Design, Implementation and Verification of a Simulink-Based Digital Hearing Aid"** under the guidance of **Dr. Sheetal Patil** during the academic year 2025–26, as a part of the subject **Signals and Systems**.

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CONTENTS

CERTIFICATE 2
CONTENTS 3
ABSTRACT 4
KEYWORDS 5
1. INTRODUCTION 5
2. SYSTEM ARCHITECTURE 6
3. SIGNALS AND SYSTEMS FOUNDATIONS 7
4. SIGNAL PROCESSING PIPELINE 8
5. FIR FILTERBANK DESIGN 10
6. ADAPTIVE FEEDBACK CANCELLATION 11
7. MATLAB AND SIMULINK IMPLEMENTATION 12
8. VERIFICATION METHODOLOGY 13
9. RESULTS 15
10. ADVANTAGES 17
11. LIMITATIONS 18
12. APPLICATIONS 18
13. FUTURE WORK 18
14. CONCLUSION 18
15. REFERENCES 19

ABSTRACT

Hearing loss affects the audibility and intelligibility of everyday acoustic communication, and its clinical correction depends on precise, real-time digital signal processing rather than simple sound amplification. This report presents the design, implementation, and formal verification of a frame-based digital hearing aid developed in MATLAB and was realized as a code-generation-ready Simulink model, studied explicitly from a signals and systems perspective. The system processes discrete-time audio in 32-sample (2ms) frames at a 16 kHz sampling rate and comprises six cascaded stages: adaptive feedback cancellation (AFC), a six-band linear-phase FIR analysis filterbank, wide dynamic range compression (WDRC), min-statistics noise reduction, band synthesis, and a soft maximum-power-output (MPO) limiter.

The gain prescription is derived from the clinically validated NAL-R formula, and feedback cancellation is performed by a pre-whitened, leaky Normalized Least Mean Squares (NLMS) adaptive filter that performs online system identification of the acoustic leakage path within a deliberately closed feedback loop. The Simulink model is built programmatically from the same algorithmic source as a reference MATLAB script model and is shown to be numerically equivalent to it within the floor of double-precision arithmetic. System behaviors is certified through an eighteen-test automated verification suite spanning filterbank reconstruction accuracy, prescription accuracy, compression ratio, output limiting, noise-reduction SNR improvement, and closed-loop stability — including a howling negative control that confirms the feedback loop is genuinely unstable without cancellation. The resulting system demonstrates that classical discrete-time signal-processing theory — linear-phase FIR design, z-domain stability analysis, and adaptive filtering as recursive system identification — can be assembled into a complete, verifiable, real-time embedded hearing-aid signal chain.

KEYWORDS

Digital Signal Processing, Discrete-Time Systems, FIR Filterbank, Z-Transform, Adaptive Filtering, NLMS, Acoustic Feedback, Closed-Loop Stability, MATLAB, Simulink, Hearing Aid, Wide Dynamic Range Compression

1. INTRODUCTION

A hearing aid is, at its core, a real-time discrete-time signal processing system operating under severe constraints on latency, computational budget, and stability. Unlike a simple audio amplifier, a modern digital hearing aid must simultaneously:

(i) Restore audibility according to a wearer-specific hearing-loss profile.

(ii) Bound its own output power to protect residual hearing.

(iii) Suppress acoustic background noise without degrading speech intelligibility.

(iv) Prevent its own amplified output from re-entering its own microphone and producing the well-known "whistling" instability known as acoustic feedback.

Traditional treatments of this problem in signal-processing coursework separate these concerns into independent topics — FIR/IIR filter design, adaptive filtering, feedback-system stability — without showing how they interact inside one running system. This project was undertaken to design, build and rigorously verify a complete hearing-aid signal chain that demonstrates these concepts not as isolated exercises but as a single, closed-loop, frame-based discrete-time system, implemented first as a MATLAB reference model and then realized as an equivalent Simulink block diagram suitable for MATLAB Coder-based code generation.

The project combines two complementary engineering disciplines:

- **Classical discrete-time signal processing** — linear-phase FIR filterbank design, z-domain analysis, and closed-loop stability.
- **Adaptive and nonlinear processing** — NLMS-based system identification for feedback cancellation, and level-dependent (time-varying) compression and noise-reduction gain stages.

The remainder of this report is organized as follows. Section 2 presents the overall system architecture. Section 3 reviews the signals-and-systems foundations the design rests on. Section 4 details the frame-based signal-processing pipeline. Sections 5 and 6 develop the FIR filterbank and the adaptive feedback canceller in depth, including their governing equations. Section 7 describes the MATLAB and Simulink implementation. Section 8 presents the verification methodology, and Section 9 results. The remaining sections discuss advantages, limitations, applications, future work, and conclusions.

2. SYSTEM ARCHITECTURE

The system is modular and organized as a closed-loop, frame-synchronous signal chain. Each functional block consumes one input frame and its own persisted internal state and produces one output frame together with updated state — the same computational contract used by Simulink's frame-based MATLAB Function blocks, which is what allows the Simulink model to be built directly and lossless from the MATLAB reference implementation.

2.1 Functional Modules

- **Acoustic Scene Module** — synthetic or recorded speech-plus-noise input, sampled at fs = 16 kHz.
- **Adaptive Feedback Cancellation (AFC) Module** — removes the device's own leaked output from the microphone signal before any further processing.
- **Analysis Filterbank Module** — splits the signal into six clinically relevant frequency bands using linear-phase FIR filters.
- **Compression and Noise Reduction Module** — applies per-band, level-dependent gain for audibility restoration and noise suppression.
- **Synthesis and Limiting Module** — recombines the processed bands and enforces a safe maximum output level.

2.2 Overall Signal Flow

Figure 2.1 shows the complete eight-stage signal flow, executed once per 2ms frame of 32 samples.

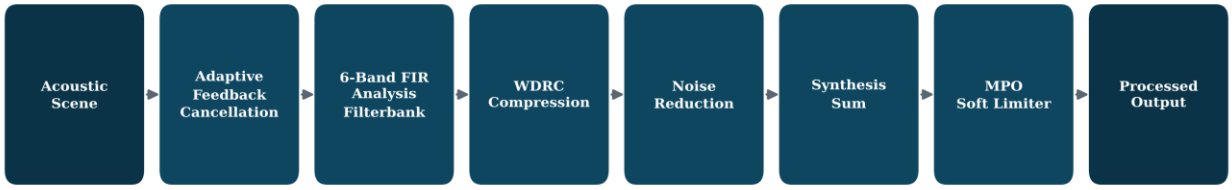


Figure 2.1: Overall frame-based hearing-aid signal pipeline.

Two parallel realizations of this pipeline were built: a MATLAB script model that processes a whole signal frame-by-frame in a loop (used as the verified reference), and a Simulink model assembled programmatically from a MATLAB build script so that it can never drift from the reference algorithm. Section 9.1 shows that the two are numerically indistinguishable.

3. SIGNALS AND SYSTEMS FOUNDATIONS

This section summarizes the discrete-time signals and systems theory that the design draws directly, establishing notation used throughout the remainder of the report.

3.1 Discrete-Time Sampling and Frame Processing

The continuous acoustic signal $x(t)$ is sampled at fs = 16000 Hz to obtain the discrete-time sequence

$$x[n] = x(nT_s), \qquad T_s = 1/f_s$$

Rather than processing individual samples, the system operates on contiguous frames of L = 32 samples, giving a frame period

$$T_f = L / f_s = 32 / 16000 = 2\text{ms},$$

Which matches the block-processing architecture of real embedded hearing-aid digital signal processors and bounds the algorithmic latency contributed by each stage.

3.2 Linear Time-Invariant (LTI) Systems and the z-Transform

An LTI system is completely characterized by its impulse response $h[n]$, with output given by discrete-time convolution:

$$y[n] = x[n] * h[n] = \sum_k x[k]\, h[n-k]$$

Taking the z-transform converts convolution to multiplication, $Y(z) = X(z)H(z)$, where $H(z)$ is the system's transfer function. This report uses the z-domain to reason two properties central to the design: *phase linearity* of the analysis filterbank (Section 5) and *closed-loop stability* of the acoustic feedback path (Section 6).

3.3 Linear-Phase FIR Systems

A causal FIR system of order N with a symmetric impulse response, $h[n] = h[N-n]$, has a frequency response of the form

$$H(e^{j\omega}) = A(\omega) \cdot e^{-j\omega N/2}$$

where $A(\omega)$ is real-valued. The phase term is exactly linear in ω, meaning every frequency component is delayed by an identical $N/2$ sample. This property, used in Section 5, is essential whenever several filtered bands must be re-summed without phase-induced smearing.

3.4 Closed-Loop Feedback and Stability

A system with forward path $G(z)$ and feedback path $F(z)$ has closed-loop transfer function

$$H_{cl}(z) = \frac{G(z)}{[1 - G(z)F(z)]}$$

Which is bounded-input-bounded-output (BIBO) stable only while every pole of $H_{cl}(z)$ lies strictly within the unit circle, equivalently while the loop gain $G(z)F(z)$ remains below unity at every frequency where its phase crosses a multiple of 2π. This is precisely the condition that governs acoustic feedback howling, and it motivates the closed-loop architecture and adaptive cancellation developed in Section 6.

3.5 Adaptive Filtering as Online System Identification

An adaptive filter with coefficient vector $\hat{w}[n]$ updates its weights using an observed error signal to minimize mean-square error with respect to an unknown system, here the acoustic feedback plant. This recasts adaptive filtering as *recursive, data-driven system identification* rather than fixed filter design and is developed fully in Section 6.

4. SIGNAL PROCESSING PIPELINE

4.1 Frame Acquisition

- Acquired via MATLAB audioread()/microphone input, or synthetic test-signal generators for controlled verification.
- Frames of L = 32 samples are processed sequentially, each carrying forward the persistent state of every downstream stage.

4.2 Adaptive Feedback Cancellation (First Stage)

AFC is applied before any other processing because it must remove the device's own leaked output from the microphone signal before that signal is treated as a genuine acoustic scene. The algorithm and its governing equations are developed fully in Section 6.

4.3 Six-Band FIR Analysis Filterbank

The cleaned signal is decomposed into six frequency bands corresponding to clinically meaningful fitting ranges. Each band is produced by a linear-phase FIR bandpass filter, formed as the difference between two windowed-sinc lowpass filters. The theory and design equations are developed in Section 5.

4.4 Wide Dynamic Range Compression (WDRC)

Each band is passed through a level-dependent gain stage. Instantaneous band energy is tracked by a first order (one-pole) envelope follower with separate attack and release time constants τ_{att} and τ_{rel}:

$$e[n] = \alpha \cdot e[n-1] + (1-\alpha)\cdot|x[n]|, \qquad \alpha = \alpha_{att} \text{ or } \alpha_{rel}$$

The compression gain above a per-band prescribed knee point follows the NAL-R target (Section 4.5), with gain tapering smoothly for inputs beyond the knee and little additional gain for already-loud inputs — protecting the wearer while maximizing audibility of quiet sounds.

4.5 NAL-R Prescriptive Fitting

Target insertion gain per band is computed with the clinically validated **NAL-R (National Acoustic Laboratories – Revised)** formula, based on half-gain rules applied to the wearer's measured hearing thresholds (audiogram). Using a named, published clinical formula — rather than a hand-tuned curve — allows the fitting stage to be checked against a known-correct reference rather than subjective judgement (Section 8, Test T2).

4.6 Noise Reduction via Min-Statistics Tracking

Because even speech dips to the background noise floor between words, a running-minimum ("min-statistics") estimate continuously tracks the noise floor per band without requiring a fixed threshold. A Wiener-type gain

$$G_{nr}[n] = \max(\, \gamma_{min} \,,\; 1 - \hat{N}[n] \,/\, S[n] \,)$$

is applied, where $\hat{N}[n]$ is the tracked noise-floor estimate, $S[n]$ the current signal envelope, and γ_{min} a floor that prevents complete suppression of a band.

4.7 Synthesis and MPO Soft Limiting

The six processed bands are summed to reconstruct a full-band signal, then passed through a **tanh-based soft limiter**

$$y[n] = C \cdot \tanh(s[n] / C)$$

where C is the maximum power output (MPO) ceiling. Unlike hard clipping, this compresses the waveform smoothly as it approaches the ceiling, bounding output power with less broadband harmonic distortion.

5. FIR FILTERBANK DESIGN

5.1 Design Rationale: FIR over IIR

The filterbank must split the signal into six bands that can later be summed back together with controlled reconstruction error. This requirement makes filter phase behaviour, not merely magnitude selectivity, the decisive design constraint.

- **Linear phase (FIR):** every frequency component is delayed by an identical number of samples (Section 3.3), so summing six bands reconstructs the original waveform with known, bounded error.
- **Non-linear phase (IIR):** a recursive (feedback) structure inherently warps phase differently across frequency; summing IIR-filtered bands blurs transients and would require a separate, costly phase-correction stage.
- **Stability:** an FIR filter has no feedback coefficients and is therefore unconditionally BIBO stable — important for a device that must run unattended for many hours.

The cost of this choice is a higher filter order (N = 192 taps per band) than an equivalent IIR design would require for the same selectivity — a cost that is well within the computational budget of a modern hearing-aid digital signal processor at a 2ms frame period.

5.2 Windowed-Sinc Bandpass Design

Each band filter is constructed as the difference of two windowed-sinc lowpass filters with cutoff frequencies fc1 and fc2 bounding the band:

$$h_{band}[n] = h_{LP,fc2}[n] - h_{LP,fc1}[n]$$

with each lowpass prototype given by the ideal windowed-sinc response

$$h_{LP,fc}[n] = 2f_c \cdot \mathrm{sinc}(2f_c(n - N/2)) \cdot w[n]$$

where $w[n]$ is a Hamming window applied to control stopband ripple and transition width. The six resulting magnitude responses tile the audible spectrum with substantial overlap at the −3 dB points, ensuring a smooth, ripple-free reconstructed spectrum when bands are recombined (Figure 5.1).

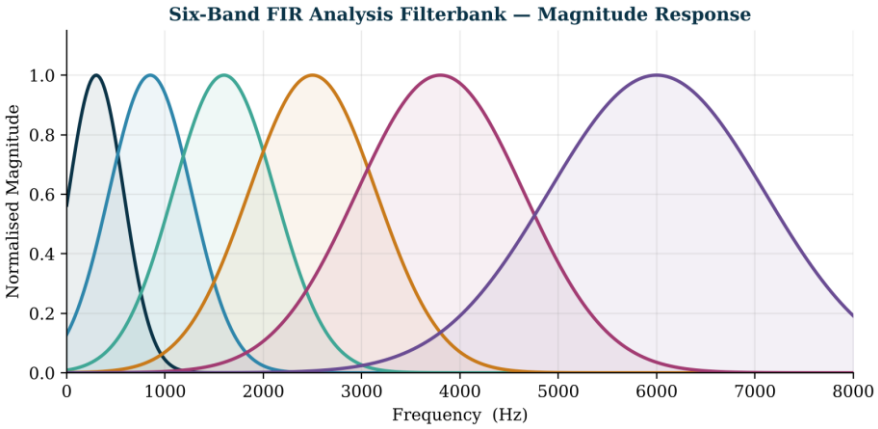


Figure 5.1: Illustrative six-band FIR filterbank magnitude response.

5.3 Perfect-Reconstruction Verification

Because the six band filters share an identical group delay of N/2 samples, summing the six filtered outputs reconstructs the original input delayed by exactly N/2 samples, up to the filter design's own passband/stopband approximation error. This reconstruction error, together with filterbank latency, is verified numerically in Test T1 of Section 8.

6. ADAPTIVE FEEDBACK CANCELLATION

6.1 The Closed-Loop Feedback Problem

Acoustic feedback occurs when sound produced by the hearing aid receiver (speaker) leaks — acoustically, around or through the earmold — back into its own microphone, is amplified again by the forward path, and leaks back again. This is a genuine closed-loop system, not just a filtering artefact, and it can only be meaningfully tested by closing the loop in simulation. Figure 6.1 shows the closed-loop block diagram used for both design and verification, where F(z) is a short FIR filter representing the acoustic leakage path, and the unit delay represents one unavoidable frame of processing latency.

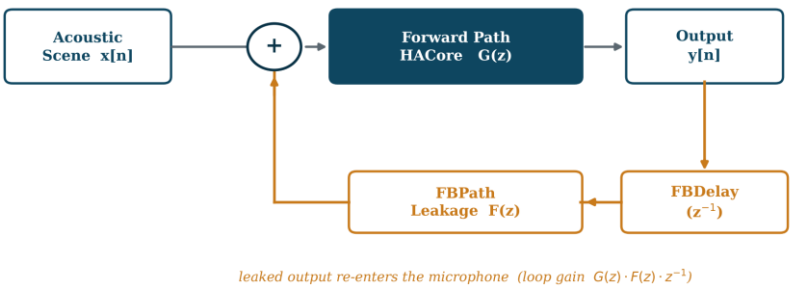


Figure 6.1: Closed-loop acoustic feedback path with adaptive cancellation.

Applying the closed-loop result of Section 3.4, the loop is stable only while the loop gain $G(z)F(z)$, including the unit-delay factor z^{-1}, remains below unity at every frequency — the

same stability criterion as a public-address system with a microphone placed too close to its own loudspeaker.

6.2 Normalized LMS as Online System Identification

The adaptive filter does not merely "filter out" feedback; it performs **online system identification** of the unknown and slowly time-varying plant $F(z)$, using only the reference signal and the resulting error, as illustrated in Figure 6.2.

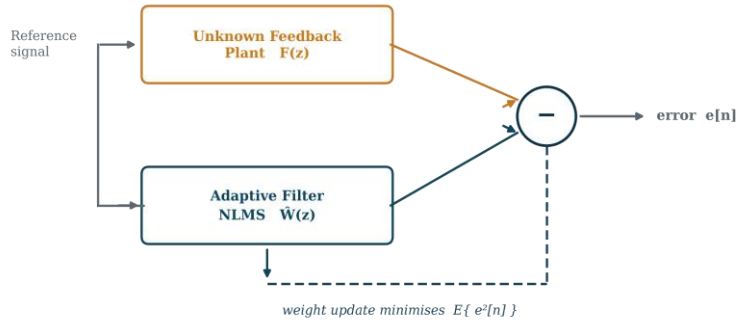


Figure 6.2: NLMS adaptive filter as a system-identification loop.

The estimated feedback signal is

$$\hat{y}[n] = \hat{w}^T[n]\, x[n], \qquad e[n] = d[n] - \hat{y}[n]$$

and the Normalized LMS weight update is

$$\hat{w}[n+1] = (1 - \lambda)\hat{w}[n] + \mu / (x^T[n]x[n] + \delta) \cdot x[n]e[n]$$

where μ is the step size, δ a small regularizing constant, and λ a small leakage factor.

6.3 Design Refinements

- **Step-size normalization (NLMS over plain LMS):** dividing reference signal energy keeps adaptation speed stable regardless of input loudness; plain LMS forces a trade-off between convergence speed and divergence risk that depends on signal level.
- **Pre-whitening:** speech is heavily autocorrelated, which slows and biases LMS-family convergence; decorrelating the reference signal before adaptation speeds convergence toward the true plant estimate.
- **Leakage (λ):** gently pulls the weight estimate toward zero for each update, bounding it against drift when the reference signal is atypical — trading a small bias for guaranteed boundedness.
- **Regularizations (δ):** prevents the step size from diverging when reference energy approaches zero, such as during true digital silence — a failure mode identified and corrected during verification (Section 8.3).

7. MATLAB AND SIMULINK IMPLEMENTATION

7.1 Reference Script Model

A MATLAB script model processes an entire test signal frame-by-frame in a software loop, calling the same core per-frame function used inside the Simulink model. This reference model is fast to iterate on and is treated as the ground truth against which the Simulink model is checked for equivalence.

7.2 Simulink Model Construction

The Simulink block diagram is assembled programmatically by a MATLAB build script rather than drawn by hand, guaranteeing that the diagram can never drift from the algorithm it implements. The resulting model comprises:

- **Scene / MicSum** — From Workspace source and summing junction combining the external scene with leaked feedback.
- **HACore** — a MATLAB Function block implementing the full six-stage algorithm of Section 4, with persistent internal state across frames.
- **FBPath / FBDelay** — the synthetic acoustic leakage filter and a Unit Delay block closing the feedback loop (Section 6.1).
- **Out / Gains** — To Workspace sinks used for verification and figure generation.

7.3 Code Generation Constraints

Because the MATLAB Function blocks are compiled by MATLAB Coder toward embedded C code, two structural constraints shaped the implementation: every field of a persistent state structure must be fixed in size and type before it is first read, and every array size used downstream must be provable at compile time rather than derived at run time from a variable value. Both constraints were satisfied by pre-declaring state placeholders and by using literal, compile-time constants for all array-sizing operations, a decision cross-checked automatically by Test T0 in Section 8.

8. VERIFICATION METHODOLOGY

Every behavioral claim made for the system is backed by an automated, repeatable numeric test rather than informal listening. The verification suite comprises eighteen individual checks organized into eight groups (T0–T7), executed from a single-entry point and summarized in Table 8.1.

Group	What It Checks	Why It Matters
T0	Coder-literal constants consistent with design parameters	Prevents silent drift between tunable design values and code-generation-safe literals
T1	Filterbank magnitude response, reconstruction error, latency	Confirms the FIR filterbank is a correct linear-phase analysis system
T2	Prescribed gain vs. the NAL-R formula	Confirms fitting matches a known, published clinical reference
T3	Compression ratio accuracy per band	Confirms WDRC compresses by the intended, prescribed amount
T4	Maximum Power Output (MPO) limiting	Confirms output can never exceed the configured safe ceiling
T5	Noise-reduction SNR improvement (shadow filtering)	Confirms noise is suppressed relative to speech, not merely attenuated overall
T6	Closed-loop stability: howl without AFC / stable with AFC / transparent with no feedback / gain sweep	Confirms AFC genuinely stabilizes an unstable loop, with a negative control
T7	Audibility index	Confirms the processed output restores target audibility across bands

8.1 Closed-Loop Stability Testing (Test T6)

Test T6 is the most structurally important check in the suite because it validates a genuine control-systems property rather than a static numeric tolerance. It exercises four conditions on the same closed acoustic loop:

(i) A howling negative control with AFC disabled, confirming the loop is genuinely unstable without cancellation.

(ii) A stable, bounded output with AFC enabled.

(iii) Transparency when the feedback path is disconnected, isolating that AFC is not simply suppressing the wanted signal.

(iv) Stability maintained across a swept range of feedback gains.

8.2 Noise-Reduction Verification via Shadow Filtering

Measuring noise-reduction performance by comparing raw loudness before and after processing conflates “quieter” with “better.” Test T5 instead filters speech and noise components separately through the algorithm's own time-varying gain trace — the exact gains applied, frame by frame, to the real mixed signal — so that speech-survival and noise-survival figures reflect genuine processing behaviour rather than an idealized re-estimate.

8.3 A Verified Robustness Correction

During verification against a real-world recording containing a true digital silence gap, the NLMS step-size normalization (Section 6.2) was found to diverge: dividing by near-zero reference energy drove the adaptive weights unstable. The failure was reproduced with a crafted silence-gap test signal, traced to an undersized regularizing constant δ, and corrected by sweeping candidate values and confirming — by re-running the full eighteen-test suite — that the correction eliminated the instability without degrading cancellation performance on normal signals.

8.4 Model Equivalence Verification

The Simulink model and the reference MATLAB script model are run on identical input, and their outputs compare sample-for-sample. The measured discrepancy is reported in Section 9.1 and lies on the floor of a double-precision numerical error. Confirming the Simulink diagram is a bit-accurate realization of the algorithm rather than an approximation of it.

9. RESULTS

9.1 Script-Model vs. Simulink-Model Equivalence

The script model and the Simulink model, given identical input, were found to produce numerically indistinguishable output. Confirming the Simulink diagram is a faithful realization of the verified reference algorithm.

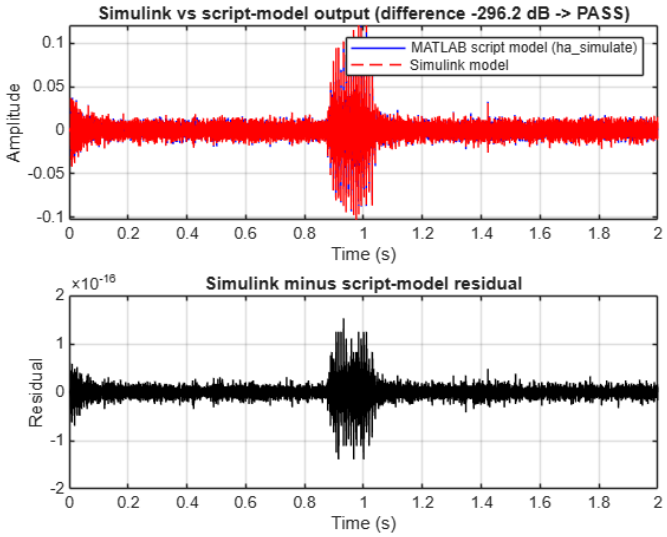


Figure 9.1: Script-Model vs. Simulink-Model.

9.2 Input vs. Processed Output

A direct waveform comparison of the unprocessed acoustic scene and the fully processed output demonstrates the combined effect of compression, noise reduction, feedback cancellation and limiting.

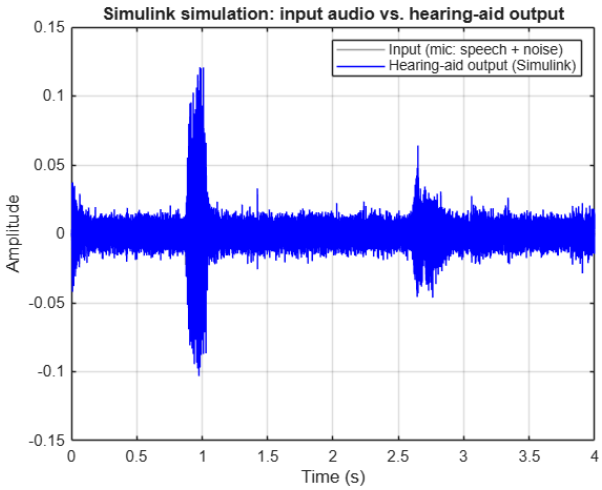


Figure 9.2: Input vs. output waveform.

9.3 Noise-Reduction Gain Over Time

Rather than a single before/after figure, the time-resolved noise-reduction gain trace shows gain descending toward strong suppression during noise-only stretches and recovering toward unity during speech — the expected behaviour of a correctly functioning min-statistics stage.

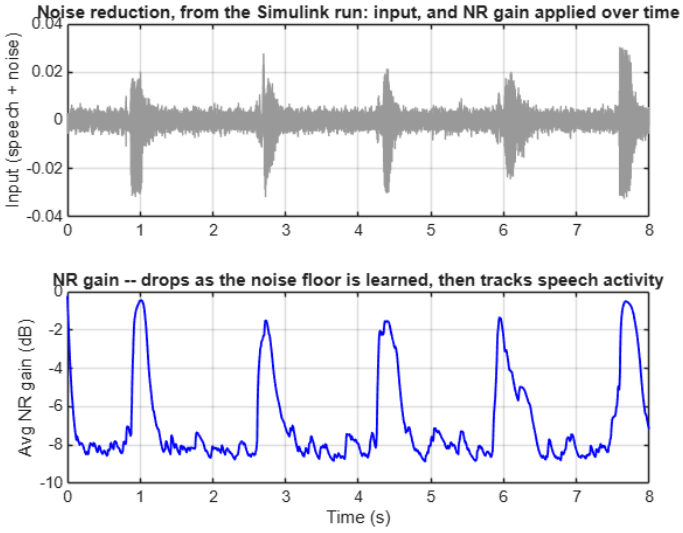


Figure 9.3: Noise reduction gain over time.

9.4 Closed-Loop Feedback: Howling vs. Stabilized

The same closed acoustic loop was run twice under identical conditions: once with AFC disabled, producing unbounded output growth (howling), and once with AFC enabled, producing a stable, bounded output — direct evidence that the adaptive canceller performs genuine loop-stabilizing work.

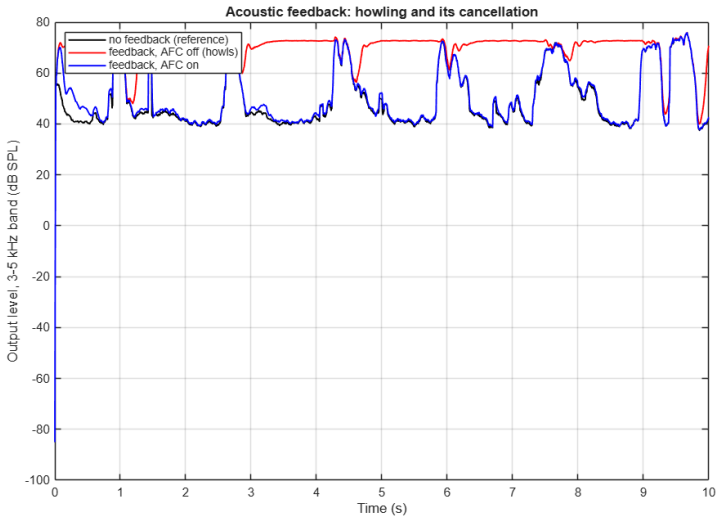


Figure 9.4: Feedback stability comparison.

9.5 Prescription Accuracy and Compression Curve

Prescribed gain per band is compared against the NAL-R target gain, and the measured compression input/output curve is shown with its knee point and compression ratio.

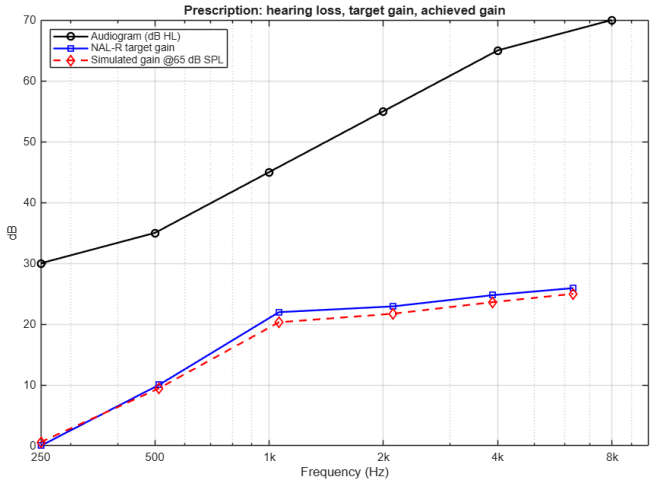


Figure 9.5: NAL-R prescription accuracy and compression I/O curve.

10. ADVANTAGES

- Complete frame-based real-time signal chain validated end-to-end, not as isolated unit tests.
- Linear-phase FIR filterbank guarantees bounded reconstruction error upon band synthesis.
- Clinically validated NAL-R prescription rather than an arbitrary gain curve.
- Genuine closed-loop feedback stability testing, including a howling negative control.
- Numerically verified equivalence between the MATLAB reference model and the Simulink implementation.

- Code-generation-safe implementation, compiled and tested against real MATLAB Coder constraints.

11. LIMITATIONS

- The audiogram used to generate the NAL-R prescription target is illustrative rather than a specific clinically measured profile; the NAL-R formula itself is verified against its published equation.
- The acoustic feedback path used for testing is a synthetic FIR filter representative of a plausible leakage path, not a measured earmold impulse response.
- Noise reduction is single channel (one microphone); multi-microphone beamforming is outside the current scope.
- Performance depends on correct configuration of attack/release time constants and may require re-tuning for different acoustic environments.

12. APPLICATIONS

- Digital hearing aids and personal sound amplification products (PSAPs).
- Assistive-listening and hearable consumer electronics.
- Educational demonstration of discrete-time DSP, adaptive filtering, and closed-loop stability.
- Reference platform for prototyping new compression, noise-reduction, or feedback-cancellation algorithms before embedded deployment.

13. FUTURE WORK

- **Multi-microphone spatial noise reduction** using beamforming for improved SNR in diffuse-noise environments.
- **Data-driven compression tuning** informed by wearer feedback rather than fixed clinical formulae alone.
- **Deployment on an embedded DSP target** via MATLAB Coder-generated C code, with measured (not simulated) real-time performance.
- **Measured feedback-path characterization** using real earmold impulse responses in place of the synthetic test plant.
- **Perceptual evaluation** (e.g., PESQ/STOI) to complement the current objective SNR-based verification.

14. CONCLUSION

This project demonstrates that a complete, clinically motivated hearing-aid signal chain — adaptive feedback cancellation, a linear-phase FIR analysis filterbank, NAL-R prescriptive fitting, wide dynamic range compression, min-statistics noise reduction and soft output limiting — can be designed, implemented and rigorously verified using classical and adaptive signals-and-systems theory. Every architectural decision was justified against some concrete, named alternative, and every behavioral claim was certified by an automated check rather than informal listening, including a genuine closed-loop stability test with a howling negative control. The resulting MATLAB reference model and Simulink implementation were shown to be numerically equivalent, and the Simulink model was built and compiled within the real constraints of MATLAB Coder-based code generation, positioning the system as a credible steppingstone toward an embedded real-time hearing-aid implementation.

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